

4 (**short**) Figure 4 shows a test rig used to determine the angle of friction between a block and a test strip. The test strip moves around the conveyor belt as shown, at an angle of θ to the horizontal. The block, which has mass m , is restrained by a linear elastic spring of stiffness k . In the absence of friction, the spring would extend by an amount e_0 . Therefore its extension Δe beyond this can be used to determine the angle of friction.

Draw a polygon of forces for the block, and derive a formula to calculate the angle of friction, ψ , in terms of θ and measurements of e_0 and Δe .

[10]

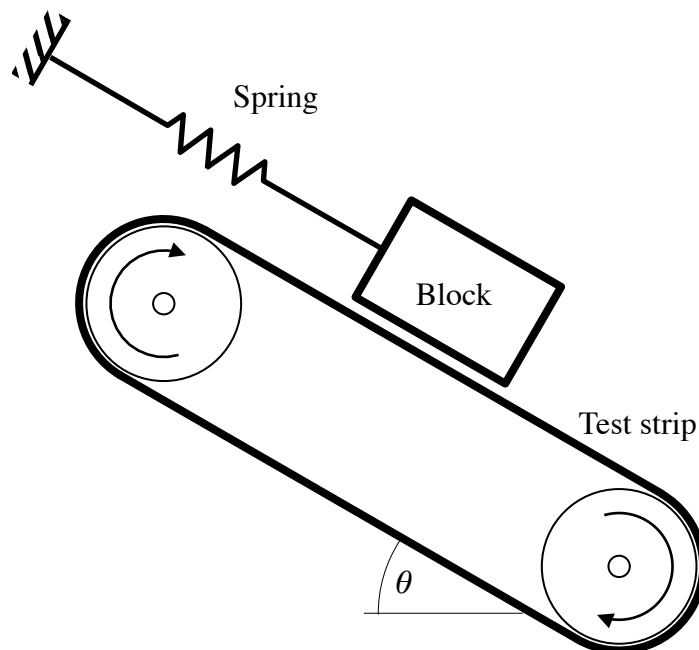


Fig. 4